

Research Paper

Server performance optimization for single-phase immersion cooling data center

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ABSTRACT

With the development of data center, air based cooling method should be updated due to the high energy consumption. Therefore, finding a high efficient cooling method becomes more and more urgent. Single-phase immersion cooling has received more attentions due to the low energy consumption. To improve the performance, more parameters and their interactive impacts should be considered. Therefore, this paper used seven key parameters to optimize server performance by using single-phase immersion cooling method, including fin height (A), fin spacing (B), thermal conductivity (C), height ratio of baffle (D), outlet area (E), inlet flowrate (F), and inlet temperature (G). The maximum temperature and pressure loss are used as the performance indicators. With the validated CFD model and response surface method (RSM), the impact of single parameter, multi-parameter, and weights of maximum temperature and pressure loss are analyzed. Results showed that, the regression equations for maximum temperature and pressure loss can well predict and optimize the server performance. In addition to the impact of single parameter, the interactive impact between height ratio of baffle and flowrate, fin spacing and flowrate, fin height and flowrate, fin spacing and height ratio of baffle, should be carefully considered when optimizing maximum temperature; the interactive impact of outlet area and flowrate should be considered when optimizing pressure loss. The weight could largely affect the optimal result and optimal value, which should be carefully designed for application. The differences in maximum temperature and pressure loss could achieve as high as 19.8 °C and 18.6 Pa when the weight of maximum temperature varied from 0.1 to 10.

1. Introduction

Data center (DC) is used to collect, store, process, and distribute massive information data, which could provide various services in different scenarios, such as entertainment, business, and other requirements [1]. With the rapid development of DCs, energy consumption receives more and more attention. It will consume 4.5 % of the global power in 2025, in which 40 % of energy is consumed by refrigeration system [2,3]. Therefore, finding a high efficient cooling method becomes more and more urgent.

There have number of works focusing on the development of new cooling methods, such as cold plate cooling [4,5], heat pipe cooling [6,7], spray cooling [8,9], two-phase immersion cooling [10], and

single-phase immersion cooling. Cold plate is embedded on central processing unit (CPU) in which the cold fluid flows to remove the heat [4]. Dede et al. [5] proposed a novel cold plate with the multi-pass branching microchannel architecture. Experimental results showed thermal resistance and pressure loss were only 0.34 (cm²·°C)/W and 19.5 kPa respectively at 100 W/cm² of heat flux. Heat pipe, considered as a material with high thermal conductivity, can be also used to take heat produced by CPU to outer space [6]. Pastukhov et al. [7] developed a miniature loop heat pipe in mobile personal computer. Results showed that the CPU temperature could maintain 83–90 °C at 25 W and 35 °C of ambient temperature. For spray cooling, the coolant is dispersed into small droplets and then sprayed on the CPU to take the heat away [8]. Salman et al. [9] found enhanced surface had higher heat transfer coefficient compared to plain surface. For two-phase immersion cooling,

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Nomenclature

a	Baffle height, mm
b	Server height, mm
C_p	Heat capability, J/(kg·°C)
e_i^f	Expected value for i KPI
g	Gravitational acceleration, m/s ²
N	The number of outlets
N	The number of KPIs
k	Thermal conductivity of solid, W/(m·°C)
P	Pressure of coolant, Pa
P_{in}	Pressure at inlet, Pa
P_{out}	Pressure at outlet, Pa
ΔP	Pressure loss, Pa
T_f	Temperature of coolant, °C
T_s	Temperature of solid, °C
u	x direction velocity, m/s
v	y direction velocity, m/s
w	z direction velocity, m/s

Z_a, Z_b Independent parameters

Greek symbol

γ_0	Regression coefficients of truncation term
γ_a	Regression coefficients of linearity term
γ_{ab}	Regression coefficients of interactive term
γ_{aa}	Regression coefficients of quadratic term
σ	Statistical error
λ	Thermal conductivity of coolant, W/(m·°C)
μ	Dynamic viscosity, Pa·s
ρ	Density, kg/m ³

Abbreviations

ANOVA	Analysis of variance
CPU	Central processing unit
DC	Data center
KPI	Key performance indicator
RSM	Response surface method

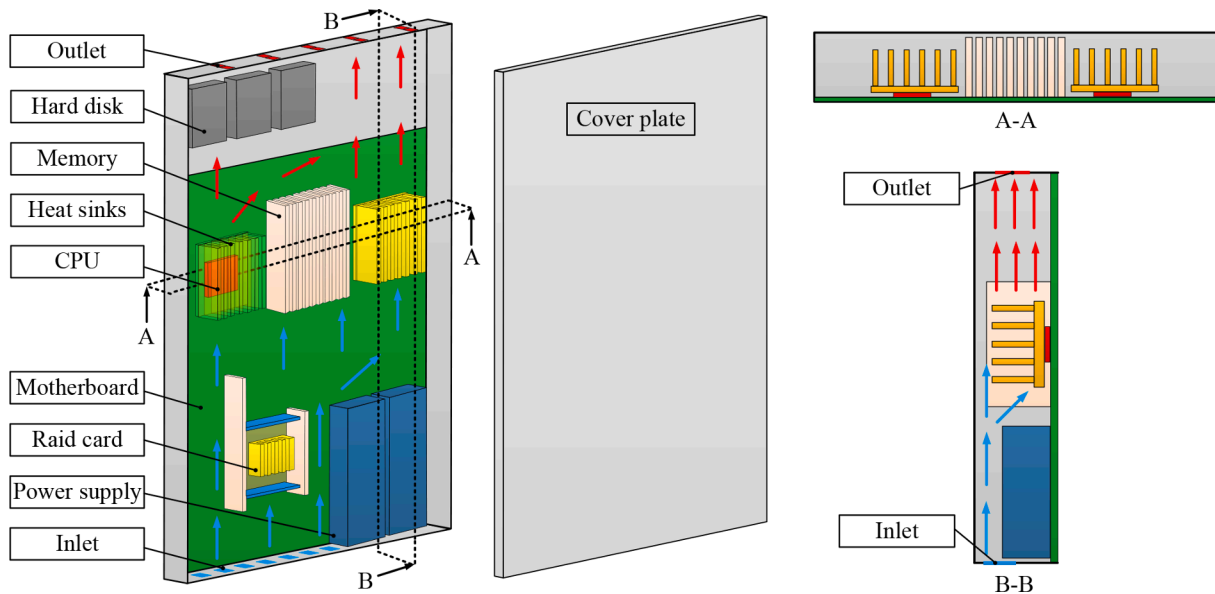


Fig. 1. Basic server structure.

Liu et al. [10] discussed the effect of different interval between servers. Results showed that a large interval would decrease the surface temperature of servers.

Compared to the conventional cooling, i.e., air cooling, single-phase immersion cooling method could save up as high as 29 % of energy consumption [11,12]. Therefore, many works are conducted to optimize its performance. Firstly, heat sink for server is a key component which affects cooling performance. Since the high viscosity of coolant, Sarangi et al. [13] suggested the heat sink for single-phase immersion cooling should have thick fins and wide fin spacing. Eiland et al. [14] found the coolant directly flowing through the heat sink could decrease the temperature by 5 °C. Secondly, the flow arrangement should be considered to make more coolant flow through the heat sink. Li et al. [15] found the server with three baffles could decrease by 0.6 °C and 1.4 °C, compared to those with one and no baffle, respectively. Luiten et al. [16] found the staggering and in-line layout had the similar cooling performance. Thirdly, the coolant has obvious impact on cooling performance. Low dynamic viscosity and low Rayleigh number had better performance

[17,18]. Nanofluid could also affect its performance. Luo et al. [19] found when the Reynold number was lower than 250, the optimal concentration was 0.3 vol%; while it was 3.7 vol% if the Reynold number was larger than 1000. Niazmand et al. [20] studied the effect of different mass concentrations for Al_2O_3 nanofluid on performance. Results showed, compared to the pure coolant, the CPU temperature decreased by 3.5 °C and pressure loss increased by only 4 % at 5 % of Al_2O_3 nanofluid. Finally, the operating parameter, such as flowrate, temperature, etc., can be investigated to improve its cooling performance. Cheng et al. [21] found CPU temperature decreased and temperature distribution became more uniform with the increase of flowrate. Bansode et al. [22] found such cooling method could keep high cooling performance under extreme condition, in which the ambient temperature ranged from −20 °C to 55 °C. Even for a prolonged time, it also showed good cooling performance [23].

Since the above factors could affect the performance, more parameters should be considered when optimizing such cooling method. For the heat sink, Li et al. [24] optimized fin thickness, channel width, fin

Table 1
Parameters of server (Basic condition).

Item	Size (mm)	Number	Power (W)	Item	Size (mm)	Number
CPU	40 × 35 × 3	2	120	Power supply	230 × 74 × 38	2
Memory	140 × 35 × 2	10	2.5	Shell	741 × 420 × 44	1
Raid card	24 × 24 × 2	1	5	Heat sink with channel	100 × 75 × 30	2
Cover plate	741 × 420 × 2	1	/	Channel width	5	/
Hard disk	100 × 74 × 7	6	/	Channel height	25	/

Table 2
Physical parameter of EC-110 [32].

Density (kg/m ³)	Dynamic viscosity (Pa·s)	Specific Heat Capacity (J/(kg·°C))	Thermal conductivity (W/(m·°C))
832.6	0.0068	2212.1	0.136

number, and substrate thickness by using orthogonal method. Results showed, under the optimal parameter, the mass of heat sink decreased by 19 %; while the temperature only increased by 4.5 % compared to the basic condition. Similar work can also be found in [25]. For operating parameter, Shah et al. [26] found the optimal inlet temperature and flowrate were respectively 40 °C and 1 L/min for 1.5 U servers. Chhertri et al. [27] proposed a regression equation for chip temperature by regarding the chip power, flowrate, and inlet temperature. Results showed the deviation was within 10 %. Gandhi et al. [28] found the optimal inlet temperature and flowrate were 45 °C and 3.8 L/min, under which the temperature could maintain around 70 °C.

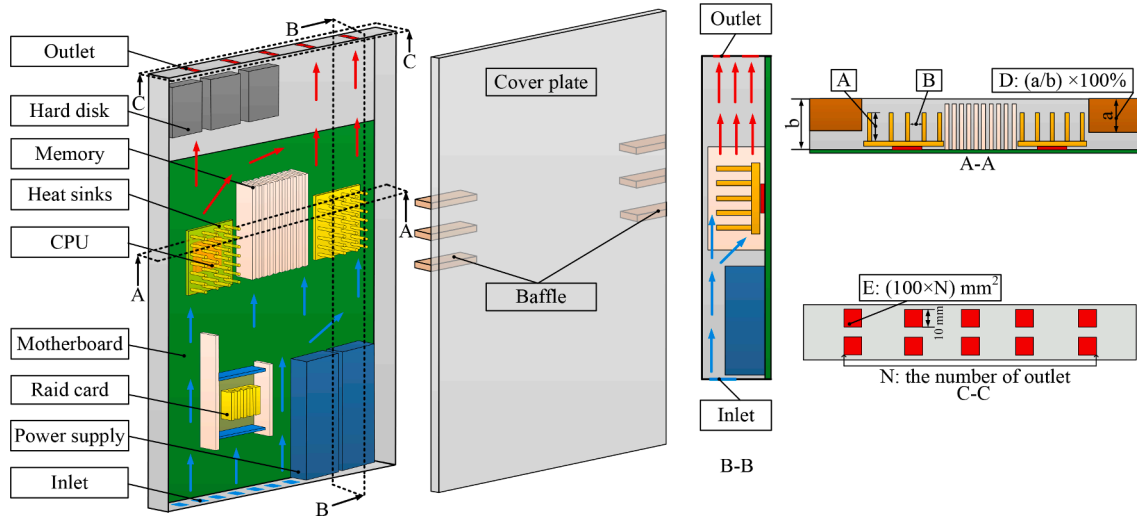


Fig. 2. Modified server structure.

Table 3
Range of key parameter [24,27].

Item	Parameters	Unit	Range	Item	Parameters	Unit	Range
A	Fin height	mm	20–30	E	Outlet area	mm ²	600–1400
B	Fin spacing	mm	3–7	F	Inlet flowrate	m ³ /h	0.1–0.5
C	Thermal conductivity	W/(m·°C)	200–400	G	Inlet temperature	°C	30–38
D	Height ratio of baffle	mm	50 %–100 %				

To better explore the impact of multi-parameter on the performance, response surface method (RSM) can be employed due to the well description of the nonlinear relationship [29,30]. Gao et al. [31] discussed the impact of nine parameters on the performance of packed bed latent heat thermal storage system. The optimal result showed the effective heat storage time, effective heat storage, and exergy efficiency decreased by 30.24 %, increased by 39.81 %, and improved by 7.50 %, respectively.

Through literature reviewed above, it is easy to found the performance of single-phase immersion cooling method could be affected by many parameters. Though there have some works optimizing such parameter, it is still not enough. Two aspects should be carefully considered: (1) More parameters, which could affect the cooling

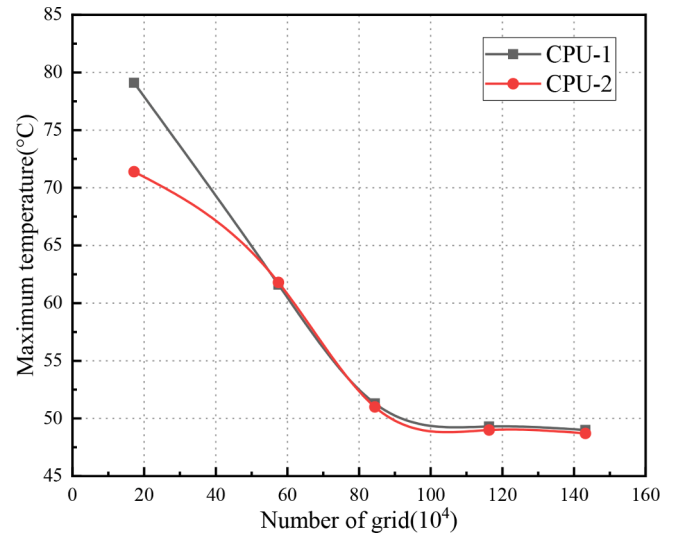
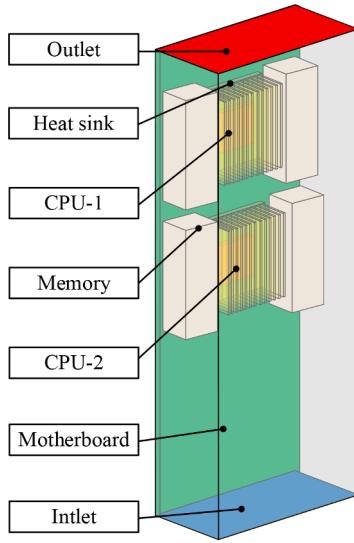
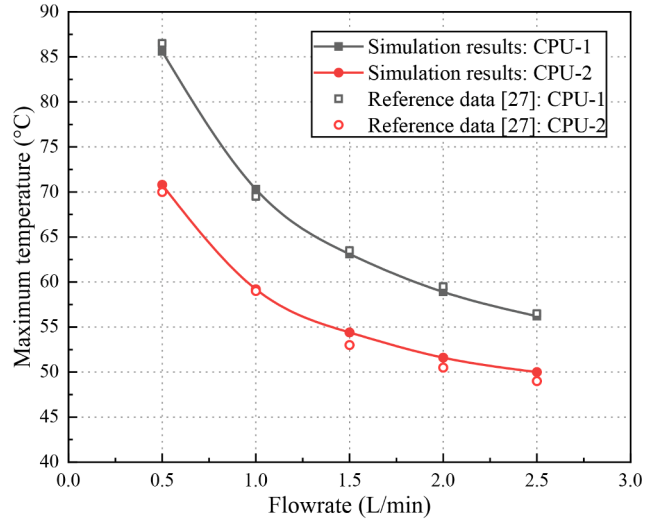


Fig. 3. Test of grid independence.



(a) Server structure



(b) Model validation

Fig. 4. Structure of server and model validation.

performance, should be considered in the optimization; (2) The interactive impact of multi-parameter also needs to be considered to further improve the performance. To fill the knowledge gap, this paper optimizes seven parameters by using validated model and RSM, which includes fin height (A) of heat sink, fin spacing (B) of heat sink, thermal conductivity (C) of heat sink, height ratio of baffle (D), outlet area (E), inlet flowrate (F) and inlet temperature (G). The maximum temperature and pressure loss are employed as key performance indicator (KPI). The result could provide guidance and general method for server optimization by using single-phase immersion cooling.

2. Server structure and research method

2.1. Model description

Server (Type: Lenovo Thinksystem SR630, 1U) is employed as basic structure to study the performance of single-phase immersion cooling system, as shown in Fig. 1. There have two CPUs, one raid card, several memories, two power supplies, and six hard disks. To simply the model, the cooling fan is dismantled and other components are ignored. The heat sink with channels is used to enhance the heat transfer. A cover plate is embedded on the server. During the operating, the coolant flows into the server and takes the heat produced by CPUs and memories. The detailed parameters for the basic structure of server can be found in Table 1.

EC-110 is used as the coolant (Produced by Engineered Fluids Company), which has the advantages of extensive material compatibility, high dielectric strength, zero environment influence and high safety capability. The physical property can be found in Table 2.

Since the coolant can flow through the other space in server, instead of the heat sink, the server structure needs to be modified to enhance the heat transfer between coolant and the CPU, as shown in Fig. 2. According to the mechanism of heat transfer and flow, three aspects are considered: (1) Heat sink: due to the large dynamic viscosity of coolant, the heat sinks on CPU are changed from channels to pin fins [20]. In addition, the fin height, the fin spacing, and the thermal conductivity of heat sink are carefully considered, i.e., parameter A, B, and C, respectively. (2) Flow arrangement: Baffles are used to improve the force flow of heat sink. The height ratio of baffle (D) and outlet area (E) are discussed, where the location of outlet is in the opposite of inlet. (3) Operating parameter: inlet flowrate and temperature are also

considered, which are marked as F and G, respectively [27]. Key parameter and its range are shown in Table 3.

2.2. Governing equation and boundary conditions

The Creo parametric and FloEFD software were used to do the simulation. Steady state is calculated to analyze the performance. The continuity equation, momentum equation, and energy equation are shown in Eqn. (1)–(3), respectively [24].

$$\frac{\partial(\rho u)}{\partial x} + \frac{\partial(\rho v)}{\partial y} + \frac{\partial(\rho w)}{\partial z} = 0 \quad (1)$$

$$\begin{cases} \frac{\partial(\rho uu)}{\partial x} + \frac{\partial(\rho uv)}{\partial y} + \frac{\partial(\rho uw)}{\partial z} = -\frac{\partial p}{\partial x} + \mu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right) \\ \frac{\partial(\rho vu)}{\partial x} + \frac{\partial(\rho vv)}{\partial y} + \frac{\partial(\rho vw)}{\partial z} = -\frac{\partial p}{\partial y} + \mu \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2} \right) \\ \frac{\partial(\rho wu)}{\partial x} + \frac{\partial(\rho wv)}{\partial y} + \frac{\partial(\rho ww)}{\partial z} = -\frac{\partial p}{\partial z} + \mu \left(\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} + \frac{\partial^2 w}{\partial z^2} \right) - \rho g \end{cases} \quad (2)$$

$$\begin{cases} \rho C_p \left(u \frac{\partial T_f}{\partial x} + v \frac{\partial T_f}{\partial y} + w \frac{\partial T_f}{\partial z} \right) = \lambda \left(\frac{\partial^2 T_f}{\partial x^2} + \frac{\partial^2 T_f}{\partial y^2} + \frac{\partial^2 T_f}{\partial z^2} \right) \\ k \left(\frac{\partial^2 T_s}{\partial x^2} + \frac{\partial^2 T_s}{\partial y^2} + \frac{\partial^2 T_s}{\partial z^2} \right) = 0 \end{cases} \quad (3)$$

where u , v , and w are the speed in x , y , and z directions; ρ is the coolant density; g is gravitational acceleration; μ is dynamic viscosity; p is pressure; C_p , T_f , λ , k , and T_s represent heat capability of coolant, temperature of coolant, thermal conductivity of coolant, thermal conductivity of solid and temperature of solid, respectively.

2.3. Key performance indicators (KPIs)

In order to accurately assess the performance of single-phase immersion cooling method, maximum temperature and pressure loss are employed. Maximum temperature of CPU should be less than 85 °C, which can be directly obtained through the simulation.

Pressure loss can be obtained through the difference between the pressure at inlet and outlet, as shown in Eqn. (4).

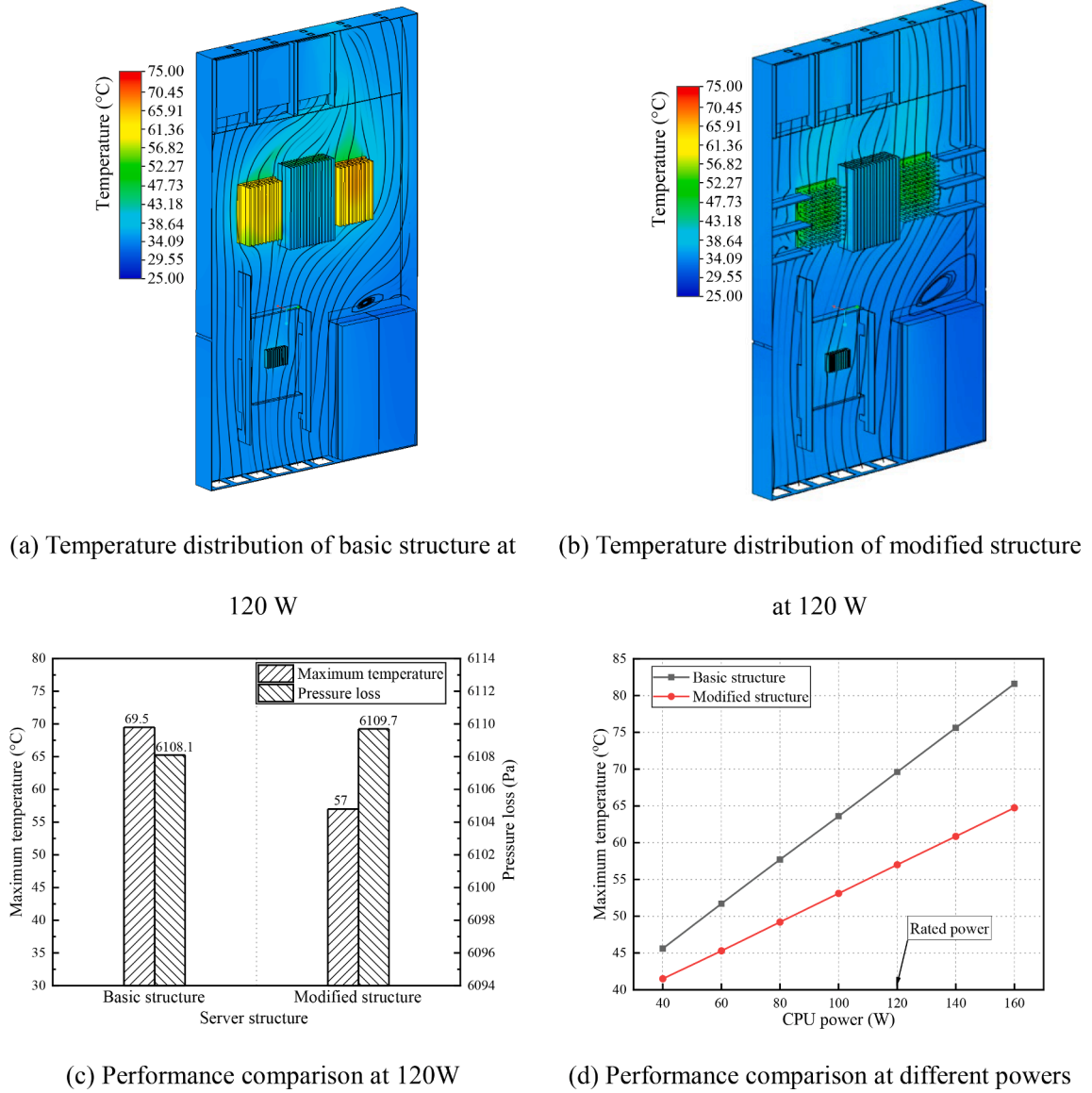


Fig. 5. Performance comparison between basic and modified structures.

Table 4
Regression model ANOVA analysis.

	F-value	Prob > F	R ²	AdjR ²	AdeP	F (Lack of fit)
Maximum temperature	78.4	<0.0001	0.9906	0.9780	36.5	1.85
Pressure loss	201.3	<0.0001	0.9942	0.9864	45.1	1.0

$$\Delta P = P_{in} - P_{out} \quad (4)$$

where P_{in} and P_{out} are pressures at inlet and outlet, respectively.

2.4. RSM introduction

RSM could illustrate the impact of the seven parameters on KPIs and obtain the optimal parameters through the multi-objective optimization. During the optimization, the cases are firstly designed based on the seven parameters, which could be found in Appendix A. Then the simulation is conducted by using the validated model. Based on the simulation result, the analysis of variance (ANOVA) and model

diagnosis are conducted to ensure the accuracy of the result. After that, the regression equation is obtained and the impacts of single parameter and multi-parameter are discussed in detail. Then, based on the regression equation, expectation function, and constraint condition (as shown in Table 3), the optimal parameter and its performance can be obtained. Finally, the impact of weights of maximum temperature and pressure loss on optimal result is discussed in detail.

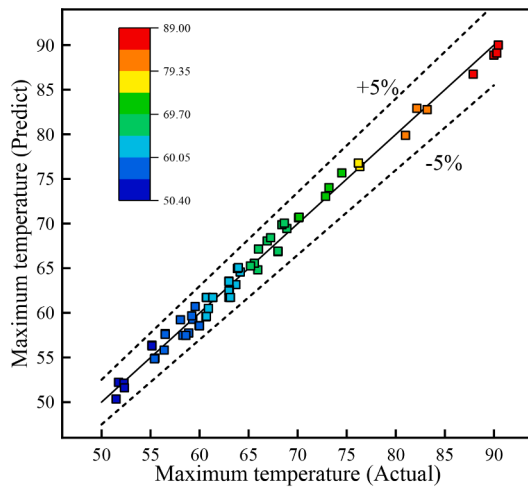
The multiple-order regression equation can be obtained by fitting the function relationship between the parameters and a KPI, as shown in Eqn. (5) [33].

$$y = \gamma_0 + \sum_{a=1}^n \gamma_a Z_a + \sum_{a=1}^n \sum_{b=a+1}^n \gamma_{ab} Z_a Z_b + \sum_{a=1}^n \gamma_{aa} Z_a^2 + \sigma \quad (5)$$

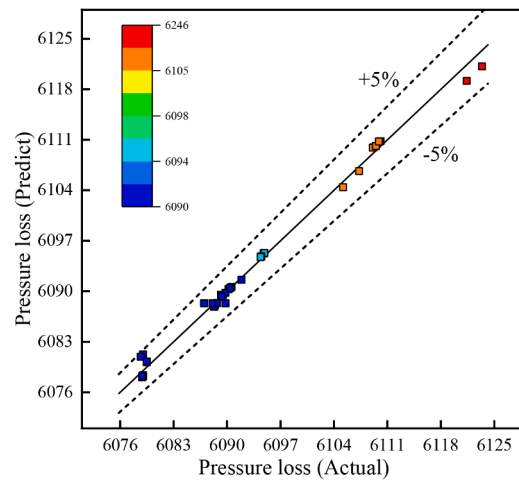
where Z_a and Z_b are independent variables (parameters), σ is the statistical error, γ_0 , γ_a , γ_{ab} , and γ_{aa} are the regression coefficients of truncation, linearity, their interactive, and quadratic term, respectively.

The expectation functions can be written as Eqn. (6) [34]:

$$E = (e_1^{r_1} \cdot e_2^{r_2} \cdot e_3^{r_3} \cdots e_n^{r_n})^{1/\sum r_i} = \left(\prod_{i=1}^n e_i^{r_i} \right)^{1/\sum r_i} \quad (6)$$



(a) Model diagnosis of maximum temperature



(b) Model diagnosis of pressure loss

Fig. 6. RSM prediction model diagnosis.

Table 5
Influence of parameters on KPIs.

	Maximum temperature	Pressure loss
Impact factor	F, D, A, C, G, B DF, BF, AF, BD	F, E, D EF
Influence	Maximum → Minimum	Maximum → Minimum

Table 6
Calculation condition of the results.

Parameter	A	B	C	D	E	F	G
A		5 mm	300 W/ (m ³ /h)	75 %	1000 mm ²	0.3 m ³ /h	34 °C
B	25 mm		300 W/ (m ³ /h)	75 %	1000 mm ²	0.3 m ³ /h	34 °C
C	25 mm	5 mm		75 %	1000 mm ²	0.3 m ³ /h	34 °C
D	25 mm	5 mm	300 W/ (m ³ /h)		1000 mm ²	0.3 m ³ /h	34 °C
E	25 mm	5 mm	300 W/ (m ³ /h)	75 %		0.3 m ³ /h	34 °C
F	25 mm	5 mm	300 W/ (m ³ /h)	75 %	1000 mm ²		34 °C
G	25 mm	5 mm	300 W/ (m ³ /h)	75 %	1000 mm ²	0.3 m ³ /h	
DF	25 mm	5 mm	300 W/ (m ³ /h)		1000 mm ²		34 °C
BF	25 mm		300 W/ (m ³ /h)	75 %	1000 mm ²		34 °C
AF		5 mm	300 W/ (m ³ /h)	75 %	1000 mm ²		34 °C
BD	25 mm		300 W/ (m ³ /h)		1000 mm ²	0.3 m ³ /h	34 °C
EF	25 mm	5 mm	300 W/ (m ³ /h)	75 %			34 °C

where e_i^n represents expected value for i KPI; n represents the number of KPIs. If any KPI or factor (parameter) exceeds its expected range, the whole function becomes zero.

2.5. Grid independence test

Fig. 3 shows the grid independence test. It was clear that, the temperature decreased sharply when the grid number increased from 17.2×10^4 to 84.4×10^4 . When it exceeds 116.3×10^4 , the variation of these

two CPUs was less than 0.5°C . Therefore, such grid number was employed to simulate the server performance.

2.6. Model validation

To validate the proposed model, the work conducted by Chhertri et al. [27] was employed. The server structure can be found in Fig. 4(a). During the operating, the inlet flowrate, inlet temperature, and heat load of CPU were from 0.5 L/min to 2.5 L/min, 40°C , and 65 W, respectively. Through Fig. 4 (b), it can be found the simulation results and reference data had the same tendency, which the temperatures of CPU-1 and CPU-2 decreased with the increase of inlet flowrate. In addition, the deviations of CPU-1 temperature and CPU-2 temperature were always less than 2°C . Therefore, the model was considered as validated and could be used to predict the server performance by using single-phase immersion cooling.

3. Results and discussion

3.1. Performance comparison between basic and modified structures

Fig. 5 shows the performance comparison between the basic and modified server structures. For basic structure, there is no baffle; and the heat sink with channels is employed. The parameters of heat sink are listed in Table 1. For modified structure, the height ratio of baffle is 75 %. The pin fin heat sink with 25 mm of fin height and 5 mm fin spacing is used. Other parameters are set as the same, which are 300 W/(m³/h) of thermal conductivity, 1000 mm² of outlet area, 0.5 m³/h of inlet flowrate and 34°C of inlet temperature. Due to the modification of heat sink and flow arrangement, more coolant could flow through the heat sink and remove the heat timely, resulting in a large decrease of maximum temperature (decrease by 12.5°C at 120 W), as shown in Fig. 5(a) and Fig. 5(b). However, such modification could slightly increase pressure loss, as shown in Fig. 5(c). Fig. 5(d) illustrates the comparison at different powers. It was clear to find the temperature difference between basic and modified structure became large at high power, indicating the modification was necessary to improve the cooling performance.

3.2. Performance optimization by RSM

3.2.1. Analysis of variance (ANOVA) and model diagnosis

In ANOVA, the F-value is used to test the accuracy of the proposed model. The Prob > F is the coefficient of significance, which should be

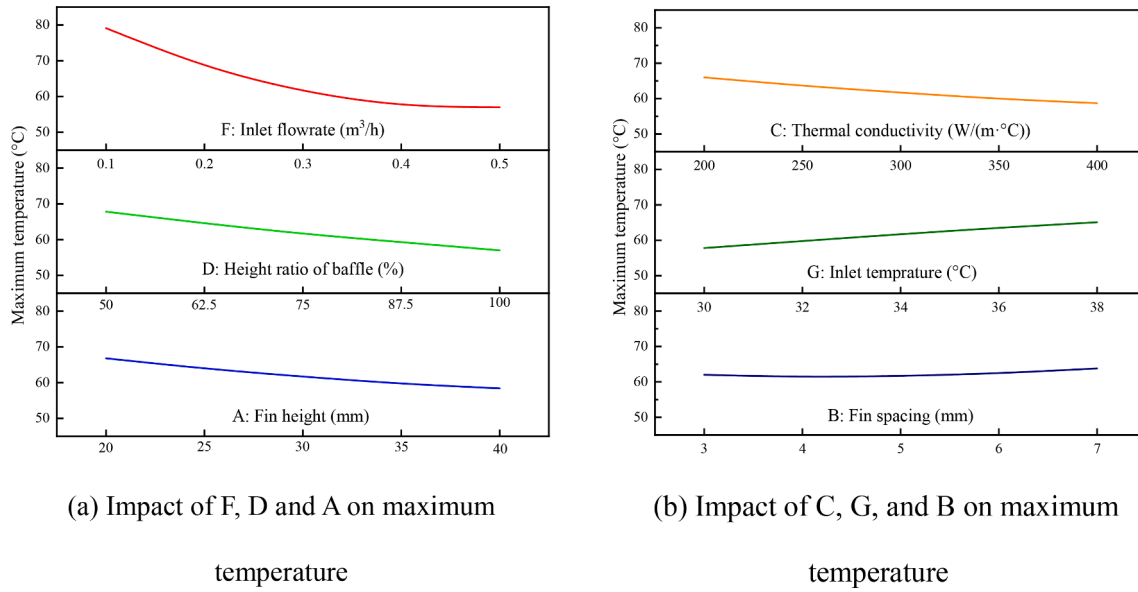


Fig. 7. Impact of parameter on maximum temperature.

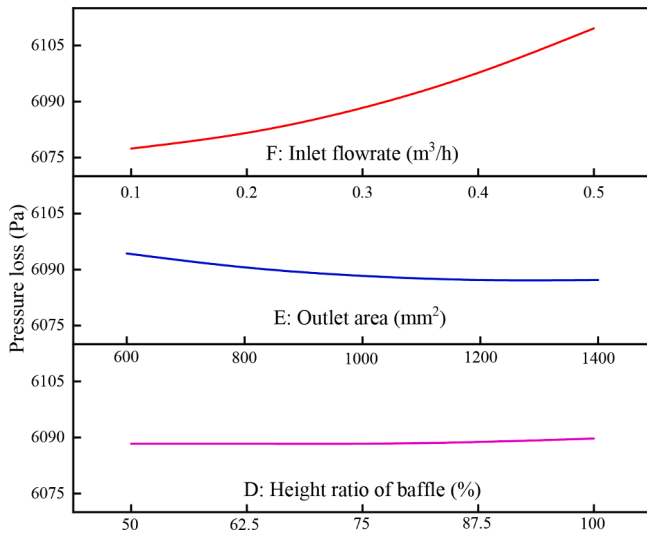


Fig. 8. Impact of parameter on pressure loss.

less than 0.05. When it is smaller than 0.0001, it means the parameters have a significant impact on the KPIs. “ R^2 ” is used to check the quality of fitting, which should be as large as possible. “Adj R^2 ” is employed to correct the accuracy of the fit, which should be as large as possible. “Adeq Precision” is the signal-to-noise ratio. If it is large than 4, the model can be used to predict the performance [31]. Through Table 4, the F-values of maximum temperature and pressure loss were 78.4 and 201.3, respectively, indicating accuracy of pressure loss was large than that of maximum temperature.

Fig. 6 shows the model diagnosis of RSM prediction. It can be found the accuracy of RSM prediction was within 5 % for maximum temperature and pressure loss. Therefore, the result of RSM was acceptable to analyze the impacts of parameter and predict the performance.

Based on RSM method, regression equations of maximum temperature and pressure loss can be obtained, as shown in Eqn. (7):

$$\begin{cases} T_{max} = 211.1 - 4.0A - 8.7B - 0.07C - 100D - 321.4F + 1.9G + 2.4AF \\ \quad + 4.1BD + 8.4BF + 84DF + 0.04A^2 + 0.3B^2 + 158.5F^2 \\ \Delta P = 6134.5 - 23.2D - 0.025E + 41F - 0.048EF + 12.7D^2 \\ \quad + 0.000015E^2 + 129.7F^2 \end{cases} \quad (7)$$

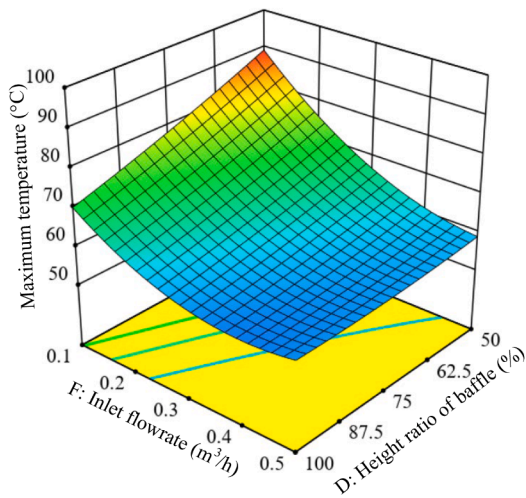
Besides, the influence of parameter on KPIs can also be obtained (see Table 5). For single parameter, F, D, A, C, G, and B would affect maximum temperature; while F, E, and D affected pressure loss. For interactive impact, DF, BF, AF, and BD would affect maximum temperature and EF affected pressure loss. It should be noted that, the interactive impact came from Eqn. (7). The impact degree of parameters can be obtained by using “Prob > F” and “F-value”, which can be found in Appendix B and Appendix C.

3.2.2. Performance analysis

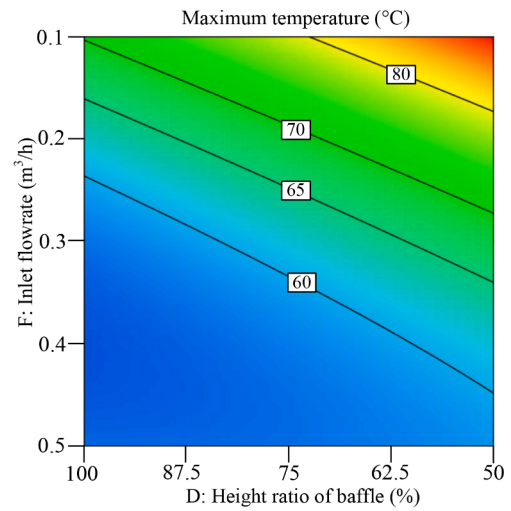
Table 6 shows the calculation condition, based on which the result is obtained. For example, when the impact of A was explained, other parameter (B, C, D, E, F, and G) can be found in the table.

Fig. 7(a) shows the impact of F, D, and A on maximum temperature. Increasing the flowrate (F) could lead to a large velocity, removing more heat from CPU timely. Through the result, maximum temperature decreased from 79.1 °C to 57 °C when the inlet flowrate increased from 0.1 m³/h to 0.5 m³/h. Moreover, the impact became small when the flowrate was large than 0.4 m³/h, indicating the rise of flowrate cannot further effectively decrease maximum temperature. Since the coolant was transferred to the heat sink at a large height ratio of baffle (D), maximum temperature linearly decreased, indicating the height ratio should be as large as possible. The similar phenomenon occurred for the fin height (A). In this scenario, the heat transfer area increased at high fin height. The linear relationship could be obtained for the impact of thermal conductivity (C) and inlet temperature (G) on maximum temperature, as shown in Fig. 7(b). It can also be found the fin spacing (B) had the slight impact on maximum temperature. During the studied range, maximum temperature only varied within 2.3 °C.

Fig. 8 shows the impact of single parameter on pressure loss. Though increasing inlet flowrate (F) could effectively decrease maximum temperature, it also increased pressure loss, especially at large flowrate. With the increase of outlet area (E), the pressure showed a decreasing tendency. Moreover, though the height ratio of baffle (D) could largely change the flow in server, its impact on pressure loss was slightly, which came from the application of heat sink.

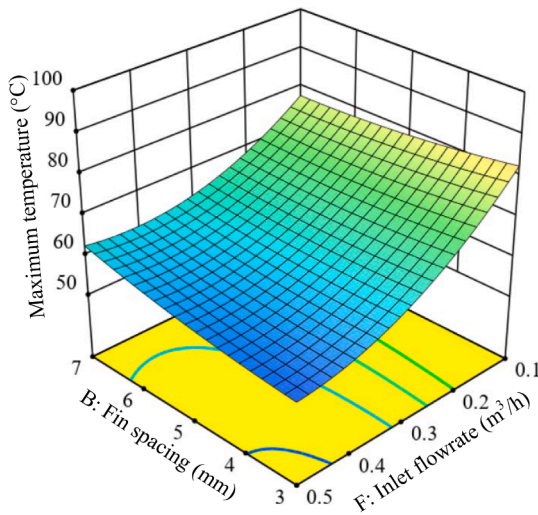


(a) Interactive impact of D and F on maximum temperature

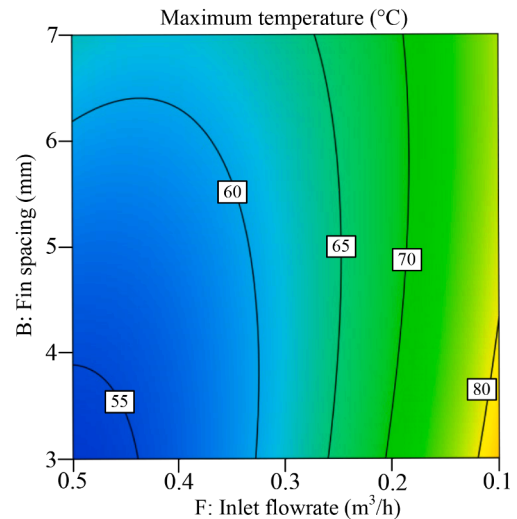


(b) Contour plot of D and F on maximum temperature

Fig. 9. Impact of D and F on maximum temperature.



(a) Interactive impact of B and F on maximum temperature



(b) Contour plot of B and F on maximum temperature

Fig. 10. Impact of B and F on maximum temperature.

Fig. 9 shows the interactive impact of D and F on maximum temperature. It can be seen that, the impact of height ratio of baffle is different under different inlet flowrates. When the flowrate was small, the increase of height ratio of baffle could obviously decrease maximum temperature. This phenomenon was mainly due to more coolant could flow through the heat sink, resulting in the decrease of maximum temperature. However, the impact of height ratio of baffle was slight under large flowrate, which was because the flowrate of heat sink was large enough to cool down the CPU, as shown in Fig. 9(a). In other words, a large height ratio of baffle is suggested to decrease the flowrate and hence, decrease the energy consumption. For instance, to keep maximum temperature at 60 °C, the flowrate was 0.44 m³/h when the height ratio of baffle was 50 %; while it was 0.22 m³/h when the ratio

was 100 %.

Fig. 10 shows the interactive impact of B and F on maximum temperature. When the flowrate was small, more coolant was needed to flow through the heat sink to take the heat away. Therefore, increasing the fin spacing was helpful to decrease maximum temperature. However, if the flowrate was large enough, more heat transfer area was required to remove the heat timely. Thus, decreasing the fin spacing could decrease maximum temperature. For example, when the fin spacing increased from 3 mm to 7 mm, maximum temperature decreased by 5 °C at 0.1 m³/h but increased by 8.4 °C at 0.5 m³/h.

Fig. 11 shows interactive impact of A and F on maximum temperature. The increase of fin height could not only enlarge the heat transfer area, but also make more coolant flow through the heat sink, resulting in

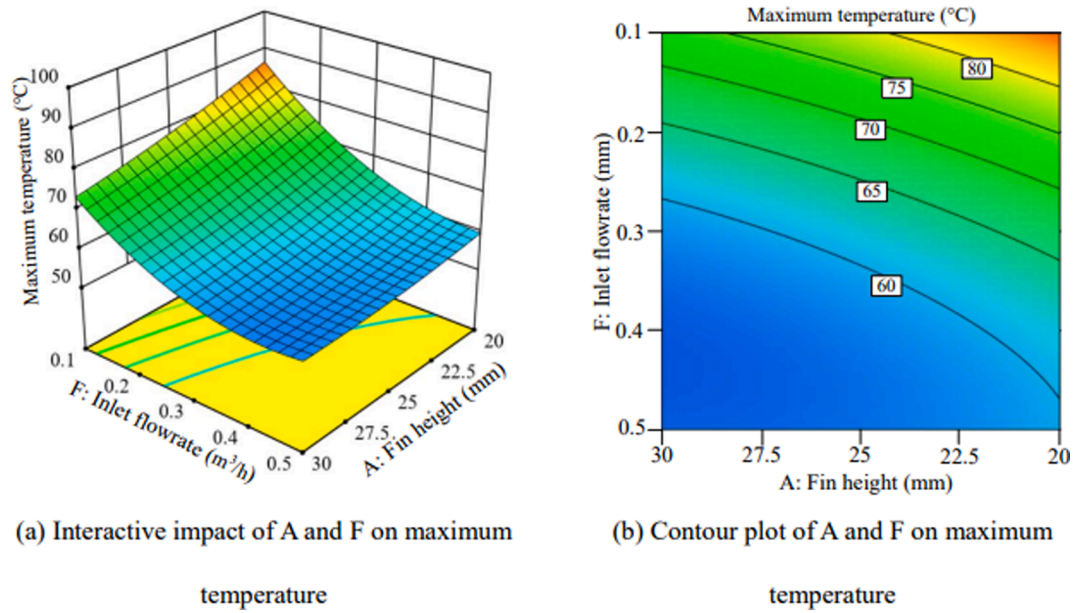


Fig. 11. Impact of A and F on maximum temperature.

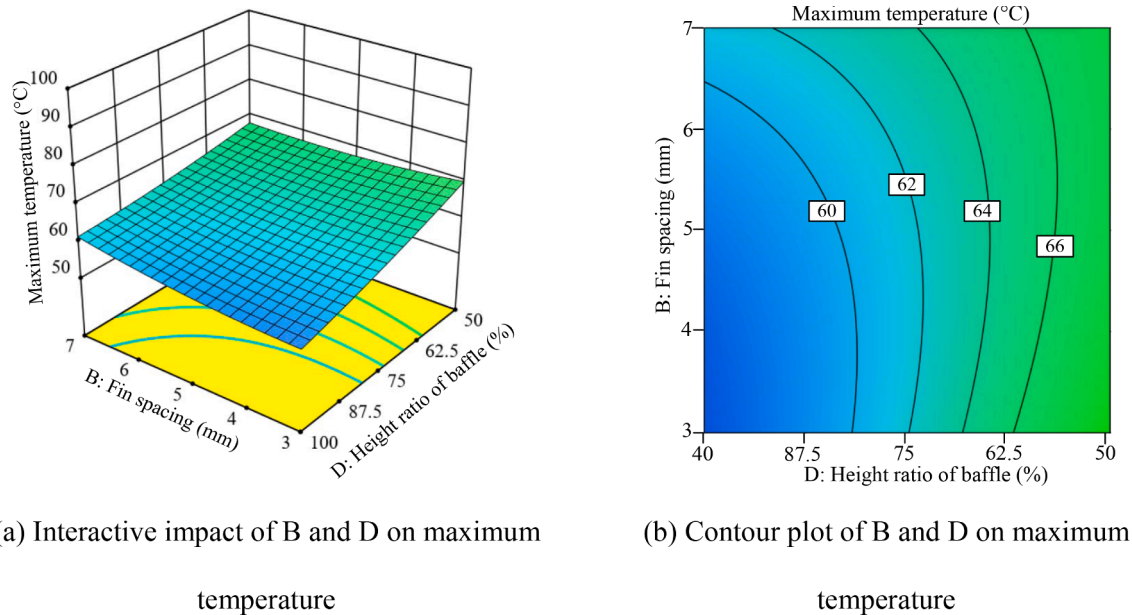


Fig. 12. Impact of B and D on maximum temperature.

the decrease of maximum temperature. Considering the impact of inlet flowrate, the impact degree of fin height was different. At $0.1 \text{ m}^3/\text{h}$ of flowrate, maximum temperature decreased by $13.1 \text{ }^\circ\text{C}$ when the fin height changed from 20 mm to 30 mm ; however, it only decreased by $3.5 \text{ }^\circ\text{C}$ at $0.5 \text{ m}^3/\text{h}$ of flowrate. If the fin height was small, more space existed in server. In such condition, increasing the flowrate could effectively decrease maximum temperature. However, when the fin height was large enough, more coolant was transferred to heat sink. And there was no need using a large flowrate. For instance, to keep maximum temperature at $60 \text{ }^\circ\text{C}$, the flowrate was $0.48 \text{ m}^3/\text{h}$ at 20 mm of fin height; while it was only $0.28 \text{ m}^3/\text{h}$ at 30 mm of fin height, as shown in Fig. 11 (b). Therefore, large fin height is suggested.

Fig. 12 shows interactive impact of B and D on maximum temperature. It can be found the impact degree of D and B was small than those in Figs. 9–11. Similarly with Fig. 10, increasing the fin spacing could

decrease maximum temperature when the height ratio of baffle was small. Oppositely, decreasing the fin spacing was helpful to decrease maximum temperature when the height ratio of baffle was large. This reason was same with that in Fig. 10. For example, when the fin spacing increased from 3 mm to 7 mm , the maximum temperature decreased by $2.1 \text{ }^\circ\text{C}$ at 50% of height ratio but increased by $5.9 \text{ }^\circ\text{C}$ at 100% of height ratio. Therefore, the fin spacing should be carefully designed regarding the height ratio of baffle and the flowrate.

Fig. 13 shows interactive impact of E and F on pressure loss. It can be found the main factor which affected pressure loss was flowrate. At large flowrate, increasing the outlet area was an effective way to decrease pressure loss. For instance, at $0.5 \text{ m}^3/\text{h}$, pressure loss decreased from 6119.3 Pa to 6104.7 Pa when the outlet area increased from 600 mm^2 to 1400 mm^2 . However, the impact of outlet area on pressure loss was slight when the flowrate was small. Overall, the outlet area should be

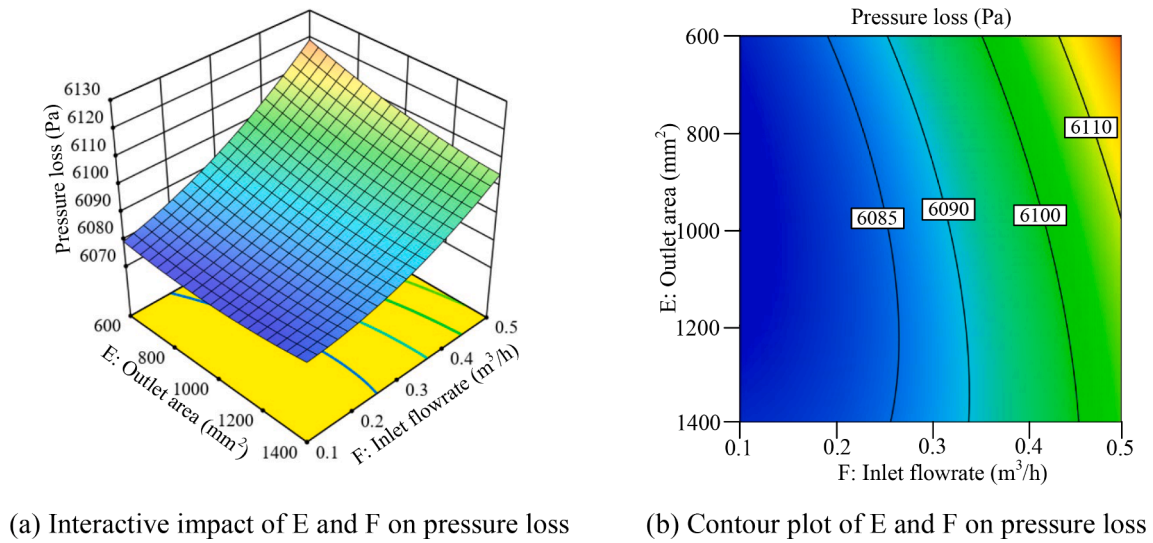
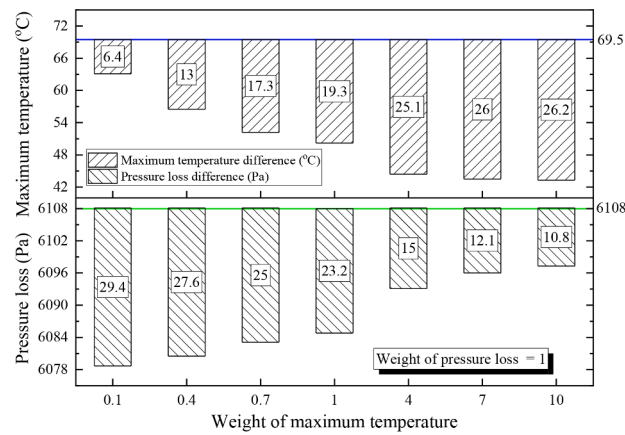
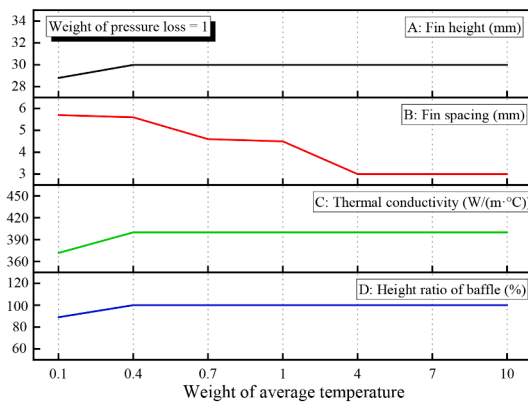


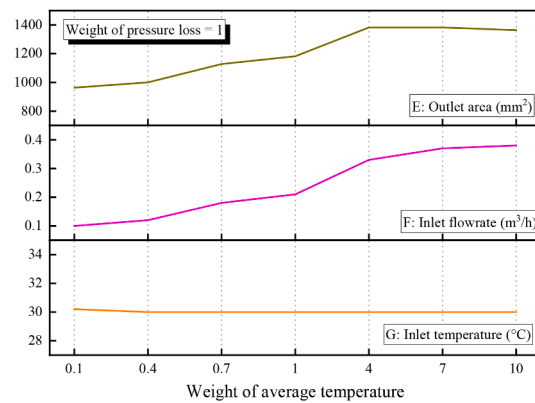
Fig. 13. Impact of E and F on pressure loss.



(a) Optimal performance under different weights of maximum temperature

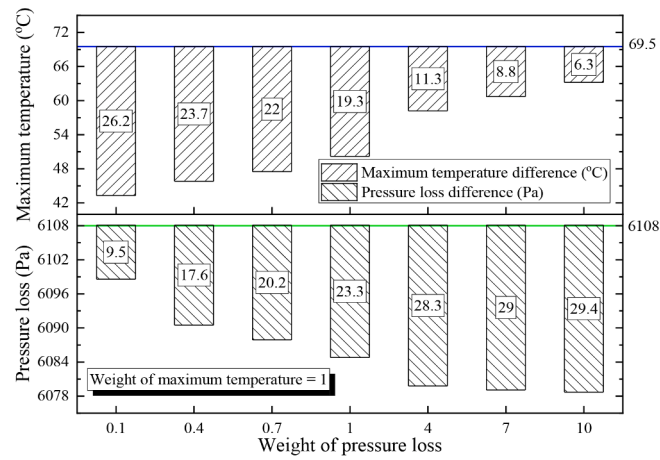


(b) Optimal parameters for A, B, C, and D under different weights of temperature

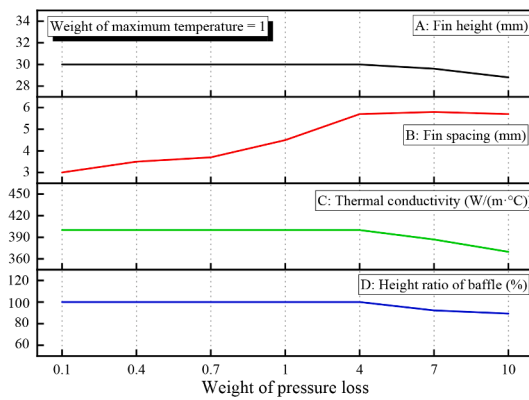


(c) Optimal parameters for E, F, and G under different weights of temperature

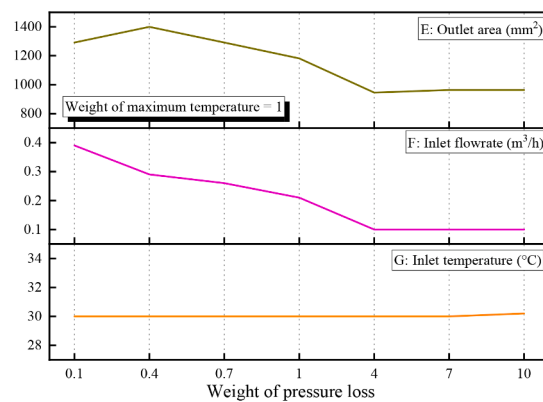
Fig. 14. Performance comparison under different weights of temperature.



(a) Optimal performance under different weights of pressure loss



(b) Optimal parameters for A, B, C, and D under different weights of pressure loss



(c) Optimal parameters for E, F, and G under different weights of pressure loss

Fig. 15. Performance comparison under different weights of pressure loss.

large as much as possible to decrease its impact on pressure loss. And the flowrate should be carefully considered since it can largely increase pressure loss.

3.2.3. Impact of weight on optimization

Since there is no data about the selection of weight in single-phase immersion cooling system, the impact of weights of maximum temperature and pressure loss are discussed in detail. Fig. 14 shows the impact of weight of maximum temperature, in which the weight of pressure loss is set as 1. It can be found that, the performance of optimal server structure was always better than that of basic server structure (the blue and green line represented the maximum temperature and pressure loss in basic structure, respectively). In addition, the weight of maximum temperature could largely affect the optimal result and optimal value for each parameter. With the increase of weight of the maximum temperature, the maximum temperature showed a decreasing tendency and pressure loss increasing tendency, as shown in Fig. 14(a). When the weight of maximum temperature varied from 0.1 to 10, the differences in maximum temperature and pressure loss could achieve as high as 19.8 °C and 18.6 Pa, respectively. Through Fig. 14 (b) and Fig. 14 (c), the optimal values for seven parameters can be obtained. For example, if the weight of the maximum temperature was 4, the optimal values for A, B, C, D, E, F, and G were 30 mm, 3 mm, 400 W/(m·°C), 100 %, 1381 mm², 0.33 m³/h and 30 °C, respectively. It should be note that, the

flowrate was 0.38 m³/h even if the weight of temperature increased to 10. The reason of this phenomenon was explained in Fig. 7, which meant the flowrate should be designed carefully regarding other parameters. The opposite tendency can be observed with the variation of weight of pressure loss, as well as the optimal values for seven parameters, as shown in Fig. 15.

3.3. Discussion

To improve the performance of single-phase immersion cooling system for data center, more parameters and their impacts should be considered, especially for the interactive impact. Though this paper is conducted based on the specific server, the proposed method can be well employed to do the analysis for other structures. Importantly, the regularity for each parameter is similar, such as the impact of flowrate, inlet temperature, and height ratio of baffle, etc., which can be directly employed. The impacts of weights of maximum temperature and pressure loss on heat transfer performance and optimal value could also provide guidance when optimizing the data center cooling in a general way.

In application, several servers would be used and arranged in a tank, in which the liquid working fluid flows through the server and removes the heat timely. In such condition, pressure loss, the flowrate distribution of each server, and the maximum temperature, should be

comprehensively considered. Therefore, the future work would focus this issue.

4. Conclusions

To improve the performance of single-phase immersion cooling data center, the server structure should be modified to enhance the heat transfer. Therefore, this paper comprehensively discussed the impact of parameter on the performance. Seven parameters, including fin height (A), fin spacing (B), thermal conductivity (C), height ratio of baffle (D), outlet area (E), inlet flowrate (F), and inlet temperature (G), are selected. The impact of single parameter and multi-parameter is conducted. Finally, the impacts of weights of maximum temperature and pressure loss are discussed in detail. Based on the result, the conclusion can be obtained:

- (1) The regression equations for maximum temperature and pressure loss can be used to predict and optimize the server performance.
- (2) The interactive impact between height ratio of baffle and flowrate, fin spacing and flowrate, fin height and flowrate, fin spacing and height ratio of baffle, should be carefully considered for maximum temperature; the interactive impact of outlet area and flowrate should be considered for pressure loss when optimizing the server structure.
- (3) The weight could largely affect the optimal result and optimal value, which should be carefully designed for application. The differences in maximum temperature and pressure loss could achieve as high as 19.8 °C and 18.6 Pa when the weight of maximum temperature varied from 0.1 to 10.

CRediT authorship contribution statement

Xueqiang Li: Methodology, Data curation, Formal analysis, Writing – review & editing. **Zhiming Xu:** Data curation, Investigation, Writing – review & editing. **Shengchun Liu:** Methodology, Supervision. **Xinyu Zhang:** Data curation, Investigation. **Haiwang Sun:** Project administration.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.applthermaleng.2023.120080>. This supplementary data includes three parts: Appendix A (case design and simulation results), Appendix B (analysis of variance on maximum temperature) and Appendix C (analysis of variance on pressure loss).

References

- [1] Q.X. Zhang, Z.H. Meng, X.W. Hong, Y.H. Zhan, J. Liu, J.B. Dong, T. Bai, J.Y. Niu, M. Jamal Deen, A survey on data center cooling systems: technology, power consumption modeling and control strategy optimization, *J. Syst. Architect.* 119 (2022), 102253, <https://doi.org/10.1016/j.sysarc.2021.102253>.
- [2] Y.N. Liu, X.X. Wei, J.Y. Xiao, Z.J. Liu, Y. Xu, Y. Tian, Energy consumption and emission mitigation prediction based on data center traffic and PUE for global data centers, *Global Energy Interconnection* 3 (3) (2020) 272–282, <https://doi.org/10.1016/j.gloi.2020.07.008>.
- [3] A.H. Khalaj, S.K. Halgamuge, A review on efficient thermal management of air-and liquid-cooled data centers: from chip to the cooling system, *Appl. Energ.* 205 (2017) 1165–1188, <https://doi.org/10.1016/j.apenergy.2017.08.037>.
- [4] Y.J. Lee, P.K. Singh, P.S. Lee, Fluid flow and heat transfer investigations on enhanced microchannel heat sink using oblique fins with parametric study, *Int. J. Heat Mass Tran.* 81 (2015) 325–336, <https://doi.org/10.1016/j.ijheatmasstransfer.2014.10.018>.
- [5] E.M. Dede, Y. Liu, Experimental and numerical investigation of a multi-pass branching microchannel heat sink, *Appl. Therm. Eng.* 55 (1–2) (2013) 51–60, <https://doi.org/10.1016/j.applthermaleng.2013.02.038>.
- [6] R.J. McGlen, R. Jachuck, S. Lin, Integrated thermal management techniques for high power electronic devices, *Appl. Therm. Eng.* 24 (8–9) (2004) 1143–1156, <https://doi.org/10.1016/j.applthermaleng.2003.12.029>.
- [7] V.G. Pastukhov, Y.F. Maidanik, C.V. Vershinin, M.A. Korukov, Miniature loop heat pipes for electronics cooling, *Appl. Therm. Eng.* 23 (9) (2003) 1125–1135, [https://doi.org/10.1016/S1359-4311\(03\)00046-2](https://doi.org/10.1016/S1359-4311(03)00046-2).
- [8] J. Kim, Spray cooling heat transfer: the state of the art, *Int. J. Heat Fluid Fl.* 28 (4) (2007) 753–767, <https://doi.org/10.1016/j.ijheatfluidflow.2006.09.003>.
- [9] A.S. Salman, N.M. Abdulrazzaq, A. Tikadar, S.K. Oudaha, J.A. Khan, Parametric study of heat transfer characteristics of enhanced surfaces in a spray cooling system: an experimental investigation, *Appl. Therm. Eng.* 183 (2021), 115824, <https://doi.org/10.1016/j.applthermaleng.2020.115824>.
- [10] C. Liu, H. Yu, Evaluation and optimization of a two-phase liquid-immersion cooling system for data centers, *Energies* 14 (5) (2021) 1395, <https://doi.org/10.3390/en14051395>.
- [11] Y.Q. Chi, J. Summers, P. Hopton, K. Deakin, A. Real, N. Kapur, H. Thompson, Case study of a data centre using enclosed, immersed, direct liquid-cooled servers, in: *Annual IEEE Semiconductor Thermal Measurement and Management Symposium*, 2014, pp. 164–173, doi: 10.1109/SEMI-THERM.2014.6892234.
- [12] P.A. Shinde, P.V. Bansode, S. Saini, R. Kasukurthy, T. Chauhan, J.M. Shah, D. Agonafer, Experimental analysis for optimization of thermal performance of a server in single phase immersion cooling, in: *Annual ASME International Electronic Packaging Technical Conference and Exhibition*, Vol. 59322, 2019, pp. V001T02A014, doi: 10.1115/IPACK2019-6590.
- [13] S. Sarangi, E.D. McAfee, D.G. Damm, J. Gullbrand, Single-phase immersion cooling performance in intel servers with immersion influenced heat sink design, in: *Annual IEEE Semiconductor Thermal Measurement, Modeling & Management Symposium*, 2022, pp. 1–5.
- [14] R. Eiland, J.E. Fernandes, M. Vallejo, A. Siddarth, D. Agonafer, V. Mulay, Thermal performance and efficiency of a mineral oil immersed server over varied environmental operating conditions, *J. Electron. Packaging* 139 (4) (2017), <https://doi.org/10.1115/1.4037526>.
- [15] B. Li, Numerical Simulation and Optimization of Single Phase Immersion Liquid Cooling for Servers, Shandong University, 2021, doi: 10.27272/d.cnki.gshdu.2021.003747.
- [16] W. Luiten, Single phase passive hydrocarbon immersion cooling of high-power ICs, in: *Annual IEEE International Workshop on Thermal Investigations of ICs and Systems*, 2021, pp. 1–6, doi: 10.1109/THERMINIC52472.2021.9626533.
- [17] S. Shao, T.Y. Gao, H.W. Yang, J. Zhao, J.J. Zhang, Evaluation of single phase immersion cooling system for high performance server chassis using dielectric coolants, in: *Annual ASME International Electronic Packaging Technical Conference and Exhibition*, Vol. 84041, 2020, pp. V001T08A010, doi: 10.1115/IPACK2020-2670.
- [18] M. Matsuoka, K. Matsuda, H. Kubo, Liquid immersion cooling technology with natural convection in data center, in: *Annual IEEE International Conference on Cloud networking*, 2017, pp. 1–7, doi: 10.1109/CloudNet.2017.8071539.
- [19] Q.Y. Luo, C.H. Wang, H.P. Wen, L.D. Liu, Research and optimization of thermophysical properties of sic oil-based nanofluids for data center immersion cooling, *Int. Commun. Heat Mass* 131 (2022), 105863, <https://doi.org/10.1016/j.icheatmasstransfer.2021.105863>.
- [20] A. Niazmand, P. Murthy, S. Saini, P. Shahi, P. Bansode, D. Agonafer, Numerical analysis of oil immersion cooling of a server using mineral oil and Al₂O₃ nanofluid, in: *Annual International Electronic Packaging Technical Conference and Exhibition*, Vol. 84041, 2020, pp. V001T08A009, doi: 10.1115/IPACK2020-2662.
- [21] C.C. Cheng, P.C. Chang, H.C. Li, F.I. Hsu, Design of a single-phase immersion cooling system through experimental and numerical analysis, *Int. J. Heat Mass Tran.* 160 (2020), 120203, <https://doi.org/10.1016/j.ijheatmasstransfer.2020.120203>.
- [22] P.V. Bansode, J.M. Shah, G. Gupta, D. Agonafer, H. Patel, D. Roe, R. Tufty, Measurement of the thermal performance of a custom-build single-phase immersion cooled server at various high and low temperatures for prolonged time, *J. Electron. Packaging* 142 (1) (2020), 011010, <https://doi.org/10.1115/1.4045156>.
- [23] N.A. Pambudi, A.M. Yusuf, A. Sarifudin, The use of single-phase immersion cooling by using two types of dielectric fluid for data center energy savings, *Energy Eng.: J. Assoc. Energy Eng.* 119(1) (2021) 275–286, doi: 10.32604/EE.2022.017356.

- [24] B. Li, Z. Cui, Q. Cao, W. Shao, Increasing efficiency of a finned heat sink using orthogonal analysis, *Energies* 14 (3) (2021) 782, <https://doi.org/10.3390/en14030782>.
- [25] J.M. Shah, R. Dandamudi, C. Bhatt, P. Rachamreddy, P. Bansode, D. Agonafer, CFD analysis of thermal shadowing and optimization of heatsinks in 3rd generation open compute server for single-phase immersion cooling, in: Annual ASME International Electronic Packaging Technical Conference and Exhibition, Vol. 59322, 2019, pp. V001T02A015, doi: 10.1115/IPACK2019-6600.
- [26] J.M. Shah, C. Bhatt, P. Rachamreddy, R. Dandamudi, S. Saini, D. Agonafer, Computational form factor study of a 3rd generation open compute server for single-phase immersion cooling, in: Annual ASME International Electronic Packaging Technical Conference and Exhibition, Vol. 59322, 2019, pp. V001T02A017, doi: 10.1115/IPACK2019-6602.
- [27] A. Chhetri, D. Kashyap, A. Mali, C. Agarwal, C. Ponraj, N. Gobinath, Numerical simulation of the single-phase immersion cooling process using a dielectric fluid in a data server, *Mater. Today: Proc.* 51 (2022) 1532–1538, <https://doi.org/10.1016/j.matpr.2021.10.325>.
- [28] D. Gandhi, U. Chowdury, T. Chauhan, P. Bansode, S. Saini, J.M. Shah, D. Agonafer, Computational analysis for thermal optimization of server for single phase immersion cooling, in: Annual ASME International Electronic Packaging Technical Conference and Exhibition, Vol. 59322, 2019, pp. V001T02A013, doi: 10.1115/IPACK2019-6587.
- [29] A. Arabameri, M.R.A. Moghaddam, A.R. Azadmehr, M.P. Shabester, Less energy and material consumption in an electrocoagulation system using AC waveform instead of DC for nickel removal: process optimization through RSM, *Chem. Eng. Process.* 174 (2022) 108869, doi: 10.1016/j.cep.2022.108869.
- [30] Y.Q. Wang, J.H. Hu, W.W. Xie, Y. Zhang, Optimization and analysis of CO₂ huff-n-puff process in shale oil reservoirs using response surface methodology (RSM), *Geofluids* 12 (2022), doi: 10.1155/2022/5927853.
- [31] L. Gao, Gegentana, Z.Z. Liu, B.Z. Sun, D.Y. Che, S.H. Li, Multi-objective optimization of thermal performance of packed bed latent heat thermal storage system based on response surface method, *Renew. Energ.* 153 (2020) 669–680, <https://doi.org/10.1016/j.renene.2020.01.157>.
- [32] ElectroCool Dielectric Coolants, <https://www.engineeredfluids.com/immersioncooling> (Accessed on Jan 2021).
- [33] N. Aslan, Y. Cebeci, Application of Box–Behnken design and response surface methodology for modeling of some Turkish coals, *Fuel* 86(1–2) (2007) 90–97, doi: 10.1016/j.fuel.2006.06.010.
- [34] W.L. Ye, P. Yang, Y.W. Liu, Multi-objective thermodynamic optimization of a free piston Stirling engine using response surface methodology, *Energ. Convers. Manage.* 176 (2018) 147–163, doi: 10.1016/j.enconman.2018.09.011.